Problem 3.1 (MATLAB)
Power spectral density

Investigate the MATLAB ***pwelch*** function for estimation of power spectral density functions. The basic syntax is ***[Sxx, f] = pwelch (x, WINDOW, NOVERLAP, NFFT, fs)***

- Find out (documentation) what is the basic methodology for ***pwelch***, i.e. how is the PSD calculation performed ?
- What is the meaning of the input parameters to the command ?
- Initially set the three middle parameters to '[ ]' (defaults)
- Define and plot various signals containing components such as, e.g., white noise, filtered white noise, deterministic periodic components, periodic components with random parameters (amplitude, frequency, phase) etc.
- Calculate and plot their autocorrelation functions
- Calculate and plot the corresponding power spectral densities and explain
- Verify by figure inspection the generic properties of power spectral density functions

Problem 3.2 (MATLAB)
LTI-filtering of random signals

Investigate filtering of random signals using LTI-filters

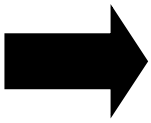
- Define an input signal x(n), e.g. white noise
- Plot the input signal and its autocorrelation function and power spectral density
- Define an arbitrary LTI-system, e.g. filter, use for example ***fir1*** or define directly a FIR impulse response for the filter
- Plot the impulse response, amplitude spectrum and magnitude squared amplitude spectrum for the filter
- Calculate the output signal y(n) using ***filter***
- Plot the output signal and its autocorrelation function and power spectral density
- Plot the input-output crosscorrelation function and the magnitude of the cross spectral density
- Verify by figure inspection the theoretical equations for the output autocorrelation function and power spectral density function and for the input-output crosscorrelation function and cross spectral density function

Problem 3.3 (MATLAB)
Coherence function for random signals

Investigate the MATLAB coherence function ***mscohere*** for calculation of frequency domain coherence between the input and output signals to a LTI-filter, where there is noise on input and/or output signal measurements.
Use the basic syntax ***[gamma_x_y_sqr_1 , f] = mscohere (x, y, [ ], [ ], [ ], fs)***

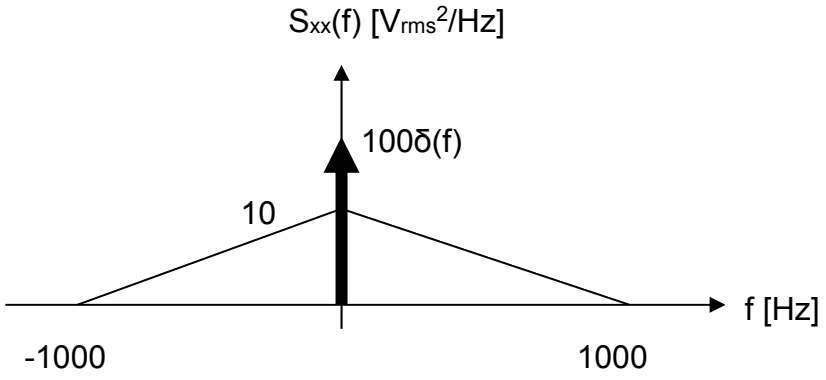
For example study the case with noise on output signal:

- Define and plot an (arbitrary) input signal x(n), e.g. white noise
- Define an (arbitrary) LTI-system, e.g. filter
- Calculate the output signal from the filter using ***filter*** and add output noise giving y(n)
- Plot the output noise and its power spectral density
- Plot the total output signal y(n)
- Calculate the coherence function between x(n) and y(n) as a function of frequency and explain
- Try different output noise signals wrt. noise power and noise amplitude spectrum

Theoretical problems 

Problem 3.4 (Theoretical)
Power spectral density and power calculation

A WSS stochastic signal $X(t)$ has the power spectral density $S_{XX}(f)$ shown below



1) Find the DC-power, indicate unit

2) Calculate the total power of the signal, indicate unit

3) Calculate the power of the signal in the bandwidth $|f| \leq 100\,[Hz]$

Problem 3.5 (Theoretical)
Cross spectral density

Assume that the WSS stochastic signals $X(t)$ and $Y(t)$ contain the components

$X(t) = A_x \cos(\omega_0 t + \theta_x) + N_x(t)$ and

$Y(t) = A_y \cos(\omega_0 t + \theta_y) + N_y(t)$

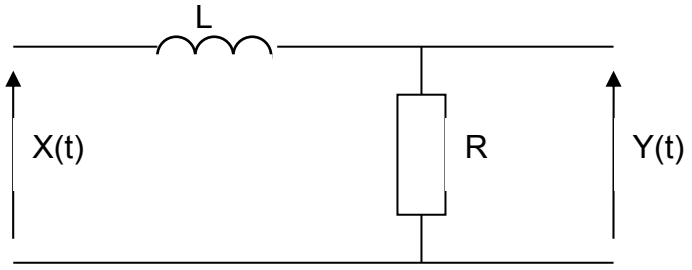
where $N_x(t)$ and $N_y(t)$ for instance can be assumed to be white noise and are mutually independent.

1) Show that the cross spectral density $S_{XY}(\omega)$ at the frequency ω_0 contains a component with amplitude $\frac{1}{2}A_x A_y$ and phase $\theta_y - \theta_x$ (first, find the cross correlation function $R_{XY}(\tau)$ and explain according to this).

2) What contains $S_{YX}(\omega)$ at the frequency ω_0?

Problem 3.6 (Theoretical)
LTI-filtering of random signals, LPF

Given the following circuit



The input-signal is the stochastic signal $X(t)$, which is gaussian, white noise

$$X(t) = N(0, \sigma_x{}^2), \quad \sigma_x{}^2 = N_0/2 \; [V_{rms}{}^2]$$

1) State $R_{XX}(\tau)$ and $S_{XX}(\omega)$ (expression and unit) and sketch

2) State the transfer function of the circuit $H(s) = \dfrac{Y(s)}{X(s)}$

   State the frequency characteristic $H(\omega)$ and sketch amplitude- and phase characteristics.
   State and sketch the impulse response $h(t)$ of the circuit.

3) Find the DC-value of the output-signal, $\mu_Y = E[Y(t)]$

4) Find the crosscorrelation function $R_{XY}(\tau)$ and sketch it. Is it symmetric wrt. the y-axis?

5) Find the cross power spectral density $S_{XY}(\omega)$ (there are (at least) 2 methods) and sketch it. Is it real or complex?

6) Find the power spectral density of the output-signal $S_{YY}(\omega)$ and sketch it.
   Is it real or complex?

7) Find the autocorrelation function of the output-signal $R_{YY}(\tau)$ (there are (at least) 2 methods) and sketch it.

8) Find the power of the output-signal P_y (there are (at least) 2 methods)

9) State the statistical distribution of the output-signal $Y(t)$

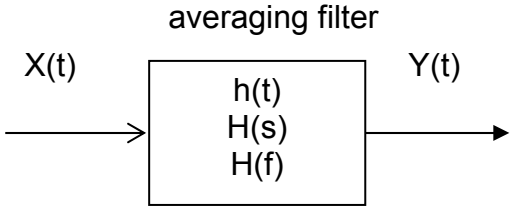
10) Now assume that the input-signal is white, uniformly distributed noise.

$$X(t) = Uniform[-a, a], \quad Var[X(t)] = \frac{(2a)^2}{12} = \sigma_x{}^2 = N_0/2 \; [V_{rms}{}^2]$$

Which of the results above will then change?

Problem 3.7 (Theoretical)
LTI-filtering of random signals, time-limited integrator

Below is shown a linear filter (LTI-system) for averaging of a stochastic input-signal.



Input-signal $X(t)$ is gaussian, white noise $X(t) = N(0, \sigma_x^2)$, $\sigma_x^2 = N_0 / 2$ $[V_{rms}^2]$

1) State $R_{XX}(\tau)$ (expression and unit) and sketch it
   State $S_{XX}(\omega)$ (expression and unit) and sketch it

The averaging filter is given at the impulse response $h(t) = \frac{1}{T}(u(t) - u(t-T))$, where T is the averaging time period.

2) State the transfer function of the filter $H(s) = \frac{Y(s)}{X(s)}$ and frequency characteristics $H(\omega)$ and sketch impulse response and amplitude- and phase characteristics

3) State an expression that shows that $Y(t)$ is an averaged version of $X(t)$, i.e. express $Y(t)$ by $X(t)$
   Apply, for instance, the impulse response (convolution) and/or the transfer function (invers Laplace)

4) Find the DC-value of the output-signal, $\mu_Y = E[Y(t)]$

5) Find the power spectral density of the output-signal $S_{YY}(\omega)$ and sketch it.

6) Find the autocorrelation function of the output-signal $R_{YY}(\tau)$

7) Find the power of the output-signal P_y in two ways, from $R_{YY}(\tau)$ and from $S_{YY}(\omega)$

8) State the statistic distribution of the output-signal $Y(t)$
   What does the averaging period T mean for the distribution and why?